Project: Worked examples
Title: Insulation thickness (ISO 12241 formula route)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

The insulation thickness on a pipe (cylinder) or a flat surface that meets a heat-flow limit, a limit on the surface temperature of a hot surface and an anti-condensation requirement for a cold one, each solved separately and the governing one reported; and for a chosen thickness the heat flow, the surface temperature and PASS/FAIL per criterion. Steady state, one layer, no thickness table of any standard. Pipe resistance per metre: R = ln(D_e/D_i)/(2π·λ) + 1/(h_se·π·D_e), D_e = D_i + 2t; with the inner film and pipe wall R_oth = 1/(h_i·π·D_p) + ln(D_i/D_p)/(2π·λ_w) added in series Flat resistance per square metre: R = t/λ + 1/h_se (+ 1/h_i + t_w/λ_w) Heat flow: q = (θ_i − θ_a)/R, positive from the process to the ambient Surface temperature: θ_se = θ_a + q/(h_se·π·D_e) (pipe), θ_a + q/h_se (flat) Surface coefficient: h_se as given, or h_se = h_c + h_r with h_r = ε·σ·(T_s² + T_a²)(T_s + T_a); h_c from free convection to air (laminar or turbulent, the larger), forced convection (cross-flow cylinder or flat plate) and their cube-law blend; solved with θ_se, which it depends on Heat-loss limit: the thickness beyond which |q| ≤ q_max; surface-temperature limit (hot): θ_se ≤ θ_se,max; anti-condensation (cold): θ_se ≥ θ_dew + margin, θ_dew entered (for example from B2) or from the Magnus relation at θ_a and RH Solve: each criterion's thickness by scanning t from 0 to 1 m for the last broken point and bisecting the bracket after it; governing t = max over the selected criteria Critical diameter: D_crit = 2λ/h_se; below it, insulation raises the heat flow until D_e passes D_crit

Ref: ISO 12241 formula route (clause numbers to be checked) — Thermal insulation for building equipment and industrial installations: calculation rules
Ref: Incropera, Fundamentals of Heat and Mass Transfer 7th ed., §7.2, §7.4, §9.6, §9.9 (to be checked) — External forced, free and mixed convection correlations
Ref: ISO 13788 cl. 6 (clause number to be checked against the owner's copy) — Magnus saturation vapour pressure (shared with B7)

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
geometry	Surface	pipe (cylinder)		assumption: A round pipe.	any
D_i	Pipe outer diameter, the inside of the insulation (pipe)	114	mm	drawing: M-PIP-021 rev B	10 ... 1000 mm
theta_i	Process (fluid) temperature	120	degC	spec: SPEC-M-12 §3: heating flow temperature	-50 ... 400 degC
theta_a	Ambient air temperature (also the radiation sink)	15	degC	spec: SPEC-M-12 §2: ambient design temperature	-20 ... 50 degC
lambda	Thermal conductivity of the insulation	0.04	W/(m*K)	assumption: Mineral wool; take the maker's value at the mean temperature (W3 material)	0.015 ... 0.2 W/(m*K)
h_mode	Where the outer surface coefficient comes from	convection + radiation (correlation)		assumption: Surface coefficient computed from the correlation.	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
h_se	Outer surface coefficient h_se (as given)	10	W/(m^2*K)	assumption: Not used: h_se is computed.	3 ... 40 W/(m^2*K)
eps	Emissivity of the outer surface (correlation)	0.1		assumption: Bright aluminium cladding; replace with the cladding maker's emissivity.	0.05 ... 1
v_wind	Air speed over the surface (correlation; 0 = still air)	0	m/s	assumption: Still air, indoors.	0 ... 20 m/s
L_surf	Height or flow length of the flat surface (correlation, flat)	1	m	assumption: Not used: a pipe.	0.1 ... 10 m
q_max_m	Heat-loss limit per metre of pipe (blank = not a criterion)	60	W/m	spec: SPEC-M-12 §5: heat loss not to exceed 60 W/m	1 ... 1000 W/m
theta_se_max	Highest surface temperature, for a hot surface (blank = not a criterion)	60	degC	spec: SPEC-M-12 §6: personnel protection, surface not above 60 °C	30 ... 70 degC
RH_a	Relative humidity of the ambient air (anti-condensation)	60	%	assumption: Not used: a hot pipe has no condensation criterion.	20 ... 95 %
t_chosen	Chosen insulation thickness (blank = report at the required thickness, no checks; with it, nothing blocks)	50	mm	spec: Pipework schedule M-SCH-021: 50 mm sections	0 ... 300 mm

3. Assumptions

A1. Steady one-dimensional heat flow through a single homogeneous layer of constant conductivity; no thermal bridges at supports, no joints, no moisture in the insulation. λ is taken at the value entered: use the value at the mean insulation temperature shown on the sheet (optionally noted in 'Temperature the conductivity applies at'). (Formula route of the standard.)

A2. With the correlation, h_se = h_c + h_r. Radiation: h_r = ε·σ·(T_s² + T_a²)(T_s + T_a) to surroundings at the ambient air temperature. Free convection to air: Holman's simplified relations at atmospheric pressure, the larger of the laminar and turbulent forms (horizontal cylinder 1.32(ΔT/D)^0.25 and 1.24ΔT^(1/3); vertical surface 1.42(ΔT/L)^0.25 and 1.31ΔT^(1/3)). Forced convection: Churchill–Bernstein for a cylinder in cross-flow, the laminar/mixed flat-plate relations along a plate; air properties from fits at the film temperature, Pr = 0.71; blended with free convection as (h_nat³ + h_forced³)^(1/3). Indicative, not the standard's own correlation. (Incropera 7th ed. §7.2, §7.4, §9.6 free convection and §9.9 mixed convection (section numbers to be checked against the owner's copy); the free-convection constants are Holman's, Heat Transfer; the standard's own coefficients are not reproduced.)

A3. The dew point is entered (for example from B2) or is the Magnus relation at θ_a and RH, the same closed form B7 uses. Below 0 °C the relation is over ice, so RH is then taken relative to ice, whereas meteorological RH is usually relative to water: enter the dew point for sub-zero air. A margin above it is the user's. (Derived; B7 shares the function.)

A4. Pipes are treated as horizontal for free convection and flat surfaces as vertical; air properties are fits

(conductivity within about 3 %, viscosity within about 4 % over 250–400 K), Pr = 0.71. (Stated.)
A5. The required thickness is the least thickness beyond which the criterion holds for every thicker layer up to 1 m, which is what matters when thin insulation on a small pipe raises the heat flow (critical diameter). (Stated.)
A6. No thickness table is used or shipped: not BS 5422, not the tables of any standard. The conductivity is the user's input, from a W3 material or the maker's data. (Addendum B §B5.)
A7. geometry (Surface) is taken as pipe (cylinder). (A round pipe.)
A8. lambda (Thermal conductivity of the insulation) is taken as 0.04 W/(m*K). (Mineral wool; take the maker's value at the mean temperature (W3 material))
A9. h_mode (Where the outer surface coefficient comes from) is taken as convection + radiation (correlation). (Surface coefficient computed from the correlation.)
A10. h_se (Outer surface coefficient h_se (as given)) is taken as 10 W/(m^2*K). (Not used: h_se is computed.)
A11. eps (Emissivity of the outer surface (correlation)) is taken as 0.1. (Bright aluminium cladding; replace with the cladding maker's emissivity.)
A12. v_wind (Air speed over the surface (correlation; 0 = still air)) is taken as 0 m/s. (Still air, indoors.)
A13. L_surf (Height or flow length of the flat surface (correlation, flat)) is taken as 1 m. (Not used: a pipe.)
A14. RH_a (Relative humidity of the ambient air (anti-condensation)) is taken as 60 %. (Not used: a hot pipe has no condensation criterion.)

4. Working

[11] t_q = t such that |q| ≤ 60 W/m for every thicker layer — ISO 12241
= scan 0 … 1000 mm, then bisection
= 23.374 mm
Required thickness, heat-loss limit

[12] t_se = t such that □_se ≤ 60 °C for every thicker layer — ISO 12241
= scan 0 … 1000 mm, then bisection
= 7.8423 mm
Required thickness, surface-temperature limit

[13] thickness = max over the selected criteria — ISO 12241
= t_q = 23.374 mm, t_se = 7.8423 mm
= 23.374 mm
Governing required thickness (heat-loss limit)

[15] D_e = D_i + 2t
= 114 mm + 2 × 50 mm
= 214 mm
Outer diameter of the insulation, chosen thickness

[16] R_ins = ln(D_e/D_i)/(2π·□) — ISO 12241
= ln(214/114)/(2π × 0.04)
= 2.5058 m·K/W
Resistance of the insulation

[17] h_c = free convection to air, blended with forced convection — Incropera 7th ed. §7, §9.6, §9.9
= horizontal cylinder, Holman; Churchill–Bernstein if the wind blows
= 3.6748 W/(m²·K)
Convective surface coefficient at the surface temperature

[18] h_r = □·□·(T_s² + T_a²)(T_s + T_a) — Incropera 7th ed. §1.2
= □ = 0.1, □_se = 27.855 °C
= 0.58007 W/(m²·K)
Radiative surface coefficient

[19] h_se = h_c + h_r — ISO 12241
= 3.6748 + 0.58007
= 4.2549 W/(m²·K)
Outer surface coefficient used

[20] R_se = 1/(h_se·π·D_e) ISO 12241
= 1/(4.2549 × π × 0.214 m)
= 0.34958 m·K/W
Resistance of the outer surface
[21] R = R_oth + R_ins + R_se ISO 12241
= 0 + 2.5058 + 0.34958
= 2.8554 m·K/W
Total resistance per metre
[22] q_per_m = |θ_i - θ_a| / R ISO 12241
= 105 K / 2.8554
= 36.773 W/m
Heat flow per metre of pipe (heat loss, magnitude)
[23] theta_se = θ_a + q/(h_se·π·D_e) ISO 12241
= 15 °C + 36.773 / 2.8606
= 27.855 degC
Outer surface temperature
[24] theta_m = (θ_i + θ_se)/2
= (120 + 27.855)/2
= 73.927 degC
Mean temperature of the insulation: take λ at this temperature
[25] D_crit = 2λ/h_se at the bare surface ISO 12241
= 2 × 0.04 / 8.1897
= 9.7683 mm
Critical diameter at the bare-surface h_se: below it, insulation raises the heat flow
[26] u_q = |q| / q_max
= |q| ≤ 60 W/m
= 0.61288
Utilisation, heat-loss limit
[27] u_se = (θ_se - θ_a)/(θ_se,max - θ_a)
= θ_se ≤ 60 °C
= 0.28567
Utilisation, surface-temperature limit

5. Results

SYMBOL	VALUE	DESCRIPTION
t_q	23.374 mm	Required thickness, heat-loss limit
u_q	0.61288	Utilisation, heat-loss limit
t_se	7.8423 mm	Required thickness, surface-temperature limit
u_se	0.28567	Utilisation, surface-temperature limit
thickness	23.374 mm	Governing required thickness
q_per_m	36.773 W/m	Heat flow per metre (heat loss)
theta_se	27.855 degC	Outer surface temperature
theta_m	73.927 degC	Mean insulation temperature
R	2.8554 m·K/W	Total resistance
D_crit	9.7683 mm	Critical diameter (at the bare-surface h_se)
h_c	3.6748 W/(m²·K)	Convective surface coefficient
h_r	0.58007 W/(m²·K)	Radiative surface coefficient

SYMBOL	VALUE	DESCRIPTION
h_se	4.2549 W/(m²·K)	Outer surface coefficient
governing	"heat-loss limit"	Governing criterion

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Chosen thickness against the heat-loss limit	u_q = 0.61288	u_q ≤ 1	0.613	ISO 12241
PASS	Chosen thickness against the surface-temperature limit	u_se = 0.28567	u_se ≤ 1	0.286	ISO 12241

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: lambda (Material properties); h_se (Outer surface coefficient); eps (Surface emissivity); q_max_m, theta_se_max (Insulation criteria). Enter each as one so its source, edition and licence travel with it.

8. Verification

Template bs.insulation@1.1.0 Content hash 38a2737abf1a41c2 Engine calc-engine/1.0.0
Known-answer cases passed 14/14 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0